

Mohit Sharma
23112062

Overview

Accurate house price prediction requires capturing both **property-level attributes** and **neighborhood-level context**. While tabular features such as floor area, room count, and coarse location provide a strong baseline, they fail to fully represent environmental characteristics like **green cover, built-up density, and urban morphology**.

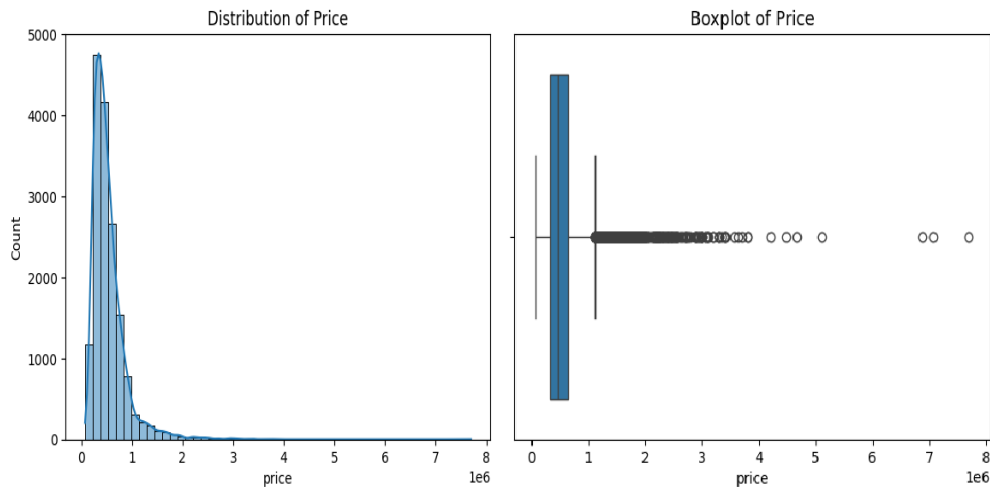
To address this limitation, this project proposes a **multimodal regression framework** that integrates structured housing data with **satellite imagery**. A tabular **XGBoost model** is first trained to establish a strong and interpretable baseline predictor. Satellite images aligned with property coordinates are then processed using a **pretrained InceptionV3 CNN** to extract high-dimensional visual embeddings representing neighborhood appearance.

Rather than directly predicting prices from images, a **residual learning strategy** is adopted. The image-based model is trained to predict the **residual errors of the tabular XGBoost model**, allowing satellite imagery to act as a **corrective signal**. This design enables the model to address **systematic biases and spatially driven errors** in the tabular predictor while preserving its stability and interpretability.

Empirical results demonstrate that the proposed **multimodal residual architecture consistently outperforms the tabular-only baseline** across standard regression metrics (RMSE, MAE, and R^2). These findings confirm that **satellite-derived visual context provides complementary information** beyond conventional structured real-estate features.

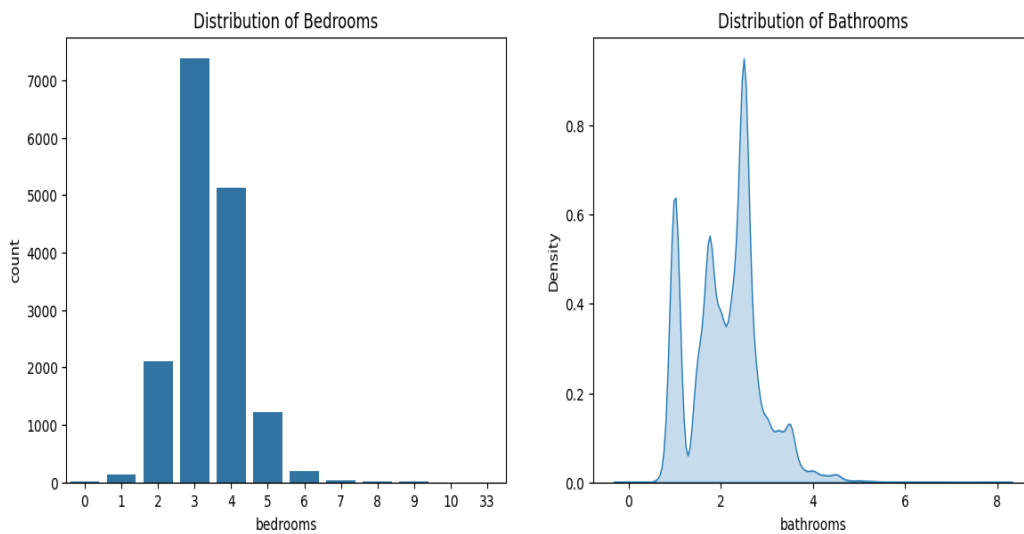
Exploratory data analysis

Figure 1: Distribution of House prices



The histogram and boxplot illustrate the distribution of house prices across the dataset, highlighting both central tendencies and extreme values. **The price distribution is strongly right-skewed**, with the majority of properties clustered in the lower price range. A **long upper tail** is evident, corresponding to a relatively small number of high-priced luxury properties. The boxplot further **confirms the presence of numerous high-value outliers**, indicating substantial variability at the upper end of the market. These characteristics suggest that price is not normally distributed and motivate the use of **robust modeling strategies (such as logarithmic transformations and tree-based methods)**.

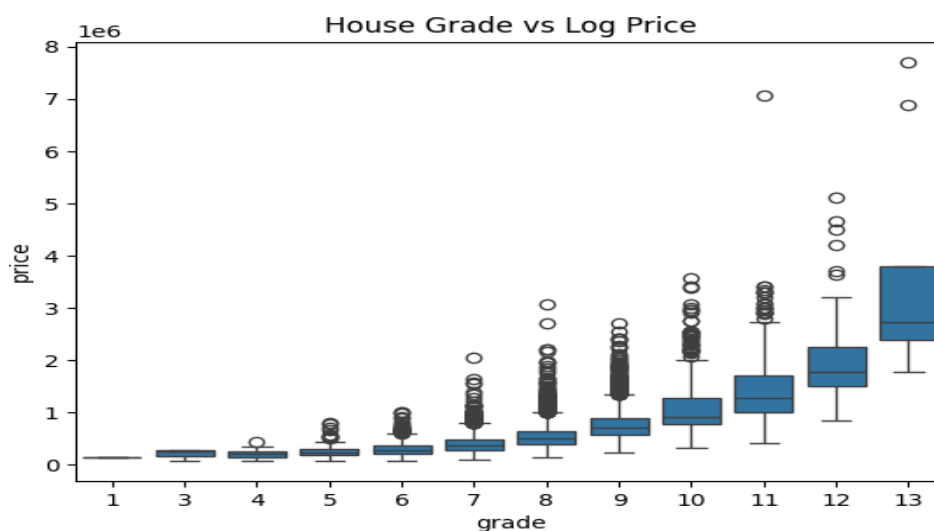
Figure 2: Distributions of Bedrooms and Bathrooms



These plots visualize the distribution of two key structural features: **the number of bedrooms and bathrooms** across properties in the dataset. Some of the features are:

Most properties have **2 to 4 bedrooms**, with a clear peak around **3 bedrooms**. Very large bedroom counts are rare and appear as extreme cases. **The distribution of bathrooms is right-skewed** and shows multiple peaks, reflecting common configurations such as **1, 1.5, 2, and 2.5 bathrooms**.

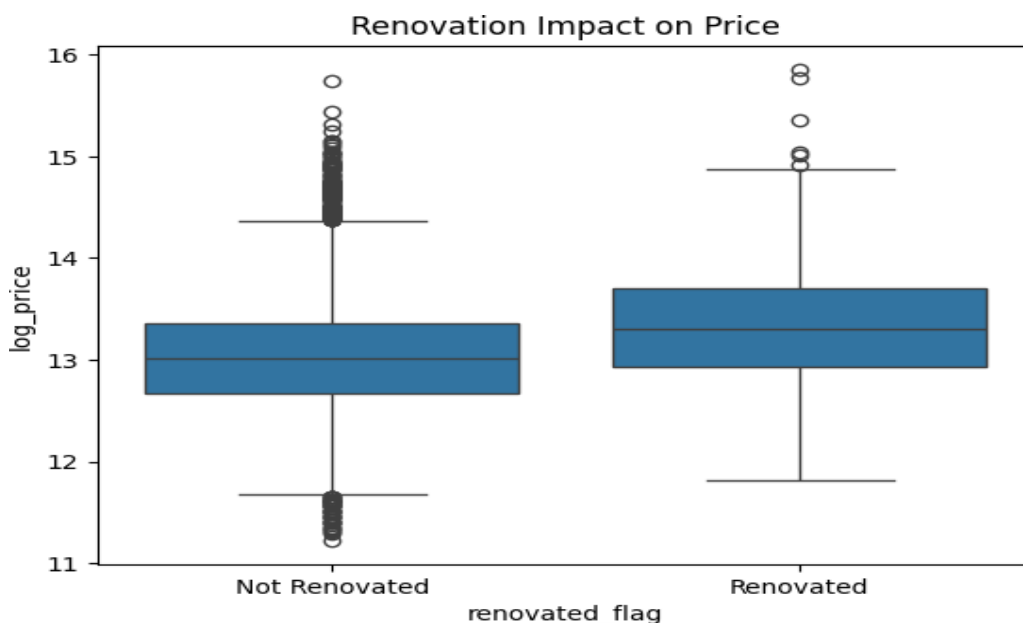
Figure 3: House Grade vs Property Price



This plot demonstrates how property prices vary across different house grade levels, where grade reflects construction quality,

architectural design, and overall build standards. Property prices show a **strong monotonic increase with grade**, indicating a clear positive relationship between build quality and market value. Lower-grade properties (grades 1–5) are associated with **consistently lower prices and limited variability**, suggesting relatively homogeneous housing quality in this segment. Mid-range grades (6–9) exhibit **increasing price dispersion**, reflecting greater heterogeneity in property features and location advantages.

Figure 4: Impact of Renovation on Property price



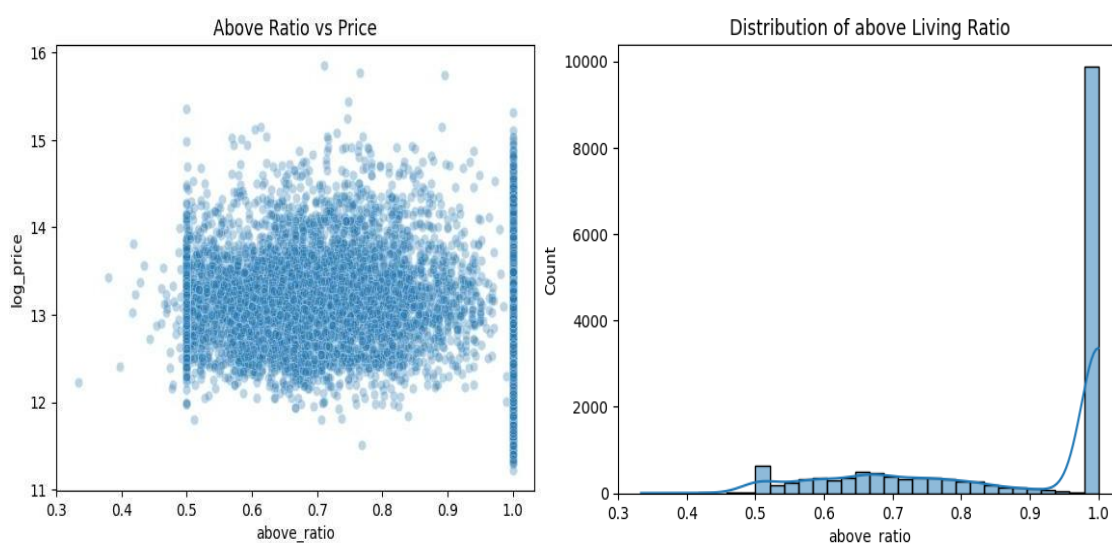
The boxplot compares the **distribution of property prices (on a logarithmic scale) for renovated and non-renovated homes**. Renovated properties exhibit a **higher median price**, indicating that renovation is generally associated with increased market value. The entire distribution for renovated homes is **shifted toward higher price ranges**, suggesting a consistent premium rather than isolated gains. While both groups show substantial variability, renovated properties tend to occupy **higher-valued segments** of the market.

Figure 5: Multicollinearity Analysis using Variance Inflation Factor (VIF)

	feature	VIF
12	long	137213.686315
11	lat	133955.154506
10	yr_renovated	16937.243693
15	renovated_flag	16935.724629
17	log_sqft_living	1772.130999
18	above_ratio	219.523831
8	grade	161.885017
7	condition	35.053544
2	bathrooms	29.354838
13	sqft_living15	26.637819

This table presents the **Variance Inflation Factor (VIF)** values for selected numerical features to assess the presence of multicollinearity among predictors. **Extremely high VIF values are observed for geographic variables (latitude and longitude), reflecting strong spatial dependence and shared location-based information.** Renovation-related features (yr_renovated and renovated_flag) also exhibit high VIF, indicating substantial redundancy between these variables.

Figure 6: Distribution and Price Association of Above-Ground Living Ratio



The histogram illustrates the **distribution of the above_ratio (sqft_above / sqft_living) feature**, while the scatter plot examines its

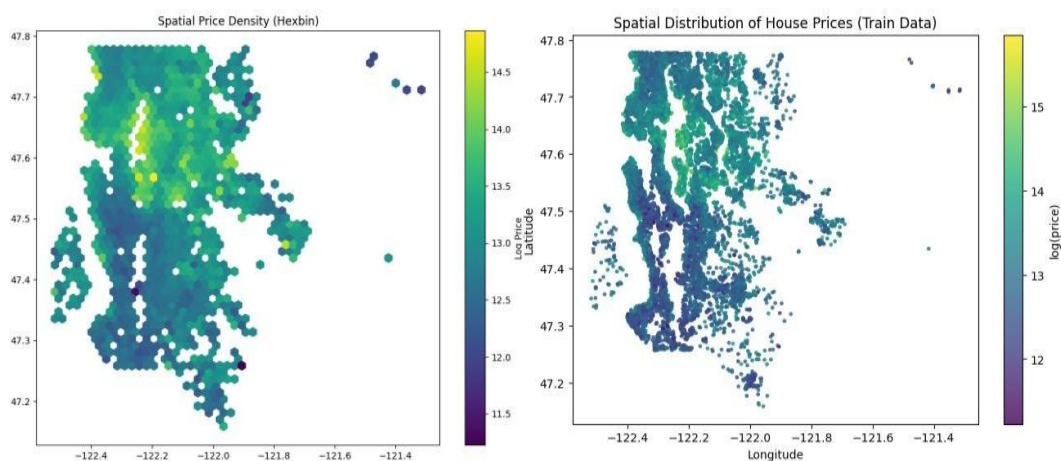
relationship with log-transformed property prices. The distribution is **strongly right-skewed**, with most properties concentrated near an **above_ratio of 1.0**, indicating that the majority of homes have nearly all living space above ground. A **secondary cluster around 0.5–0.6** suggests the presence of properties with substantial basement or below-grade living areas. The scatter plot reveals **no clear monotonic relationship** between above_ratio and log price, with prices widely dispersed across the observed ratio range.

Figure 7: Qualitative Analysis of Neighborhood Context Across Price Segments



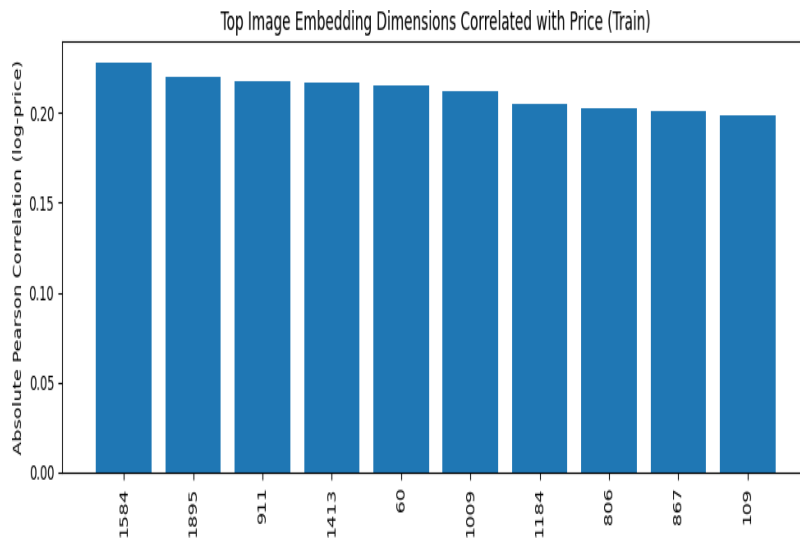
The image grids display randomly sampled **satellite views of properties grouped into low, mid, and high price buckets**, enabling qualitative comparison of neighborhood characteristics across value segments. High-priced properties are typically located in neighborhoods with **greater visible green cover**, including dense tree canopies and landscaped surroundings. These areas also tend to exhibit **lower residential density**, with houses more widely spaced and situated on larger plots rather than tightly packed clusters. Additionally, road networks in high-value regions often appear **wider and more structured**, reflecting planned urban layouts.

Figure 8: Spatial distribution and smoothed density of log(price).



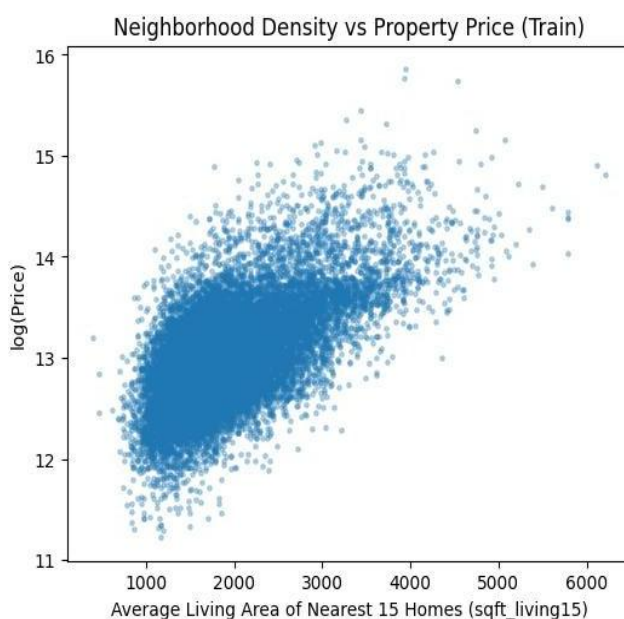
The scatter plot visualizes individual properties by latitude and longitude, with color intensity representing log-transformed sale prices, enabling detailed examination of spatial price variation. Property values exhibit **strong geographic clustering**, with contiguous corridors of higher-priced homes concentrated in central and northern parts of the region. In contrast, **lower-priced properties dominate peripheral and banded areas**, indicating systematic spatial segmentation of the housing market. The accompanying hexbin representation reduces local noise and highlights **broad high-value clusters**, suggesting that neighborhood-level spatial effects play a significant role in price formation beyond individual property characteristics.

Figure 9: Correlation Between Image Embedding Dimensions and House Prices



The bar chart displays the **absolute Pearson correlation between selected InceptionV3 embedding dimensions and log-transformed property prices**, illustrating which latent visual features are most strongly associated with property value. The most informative embedding dimensions exhibit **moderate correlations in the range of approximately 0.20–0.23**, indicating that certain visual patterns captured by the CNN consistently co-vary with house prices. The gradual decline in correlation across the top-ranked dimensions suggests that **price-relevant visual information is distributed across multiple embedding channels**, rather than dominated by a single feature.

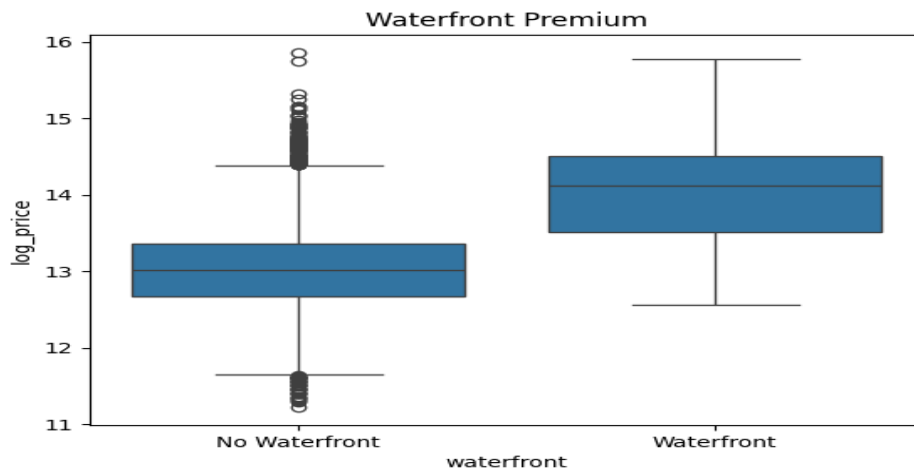
Figure 10: Neighborhood Scale vs Property Price



The scatter plot illustrates the **relationship between log-transformed**

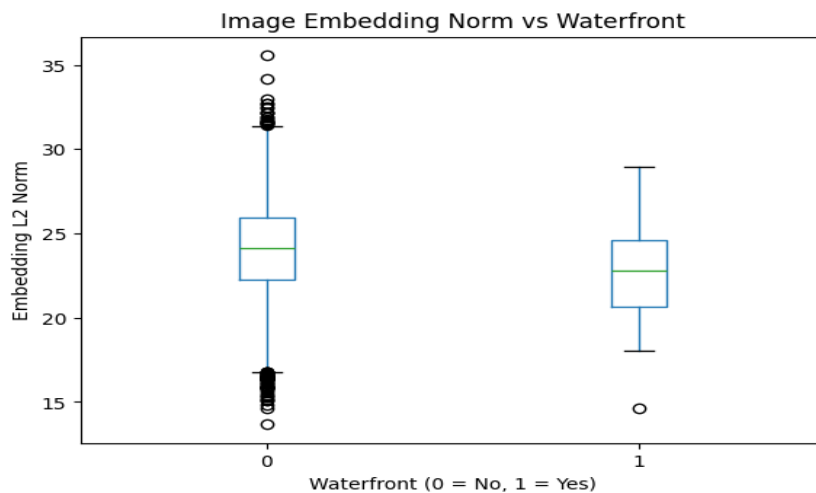
property prices and sqft_living15, which represents the average living area of the 15 nearest neighboring homes and serves as a proxy for neighborhood housing scale and density. A **clear positive association** is observed, with higher neighborhood average living areas generally corresponding to more expensive properties.

Figure 11: Waterfront label vs log price



The boxplot **compares log-transformed property prices for waterfront and non-waterfront** homes. Waterfront properties exhibit a substantially higher median price, reflecting a clear valuation premium associated with water access. The **entire price distribution for waterfront homes is shifted upward**, indicating that this premium is consistent across the market rather than driven by a few extreme cases. In contrast, **non-waterfront properties display greater dispersion and a higher concentration of lower-priced outliers**, highlighting wider variability in non-premium locations.

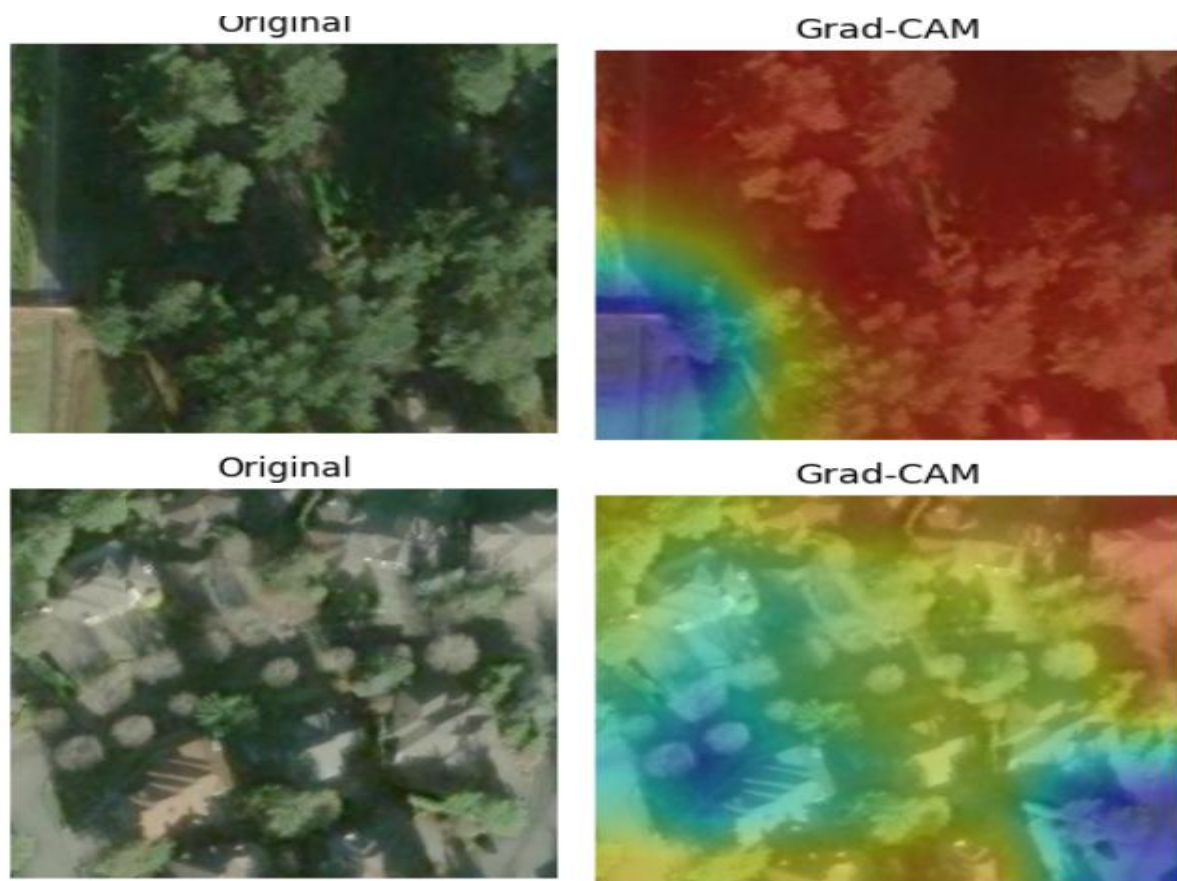
Figure 12: Image Embedding Norm by Waterfront Status



The boxplot compares log-transformed property prices for waterfront and non-waterfront homes. Waterfront properties exhibit a **substantially higher median price**, reflecting a clear valuation premium associated with water access. The entire price distribution for waterfront homes is **shifted upward**, indicating that this premium is consistent across the market rather than driven by a few extreme cases. In contrast, non-waterfront properties display **greater dispersion and a higher concentration of lower-priced outliers**, highlighting wider variability in non-premium locations. Overall, waterfront access emerges as a **strong and robust predictor** of property value.

Financial/ Visual insights

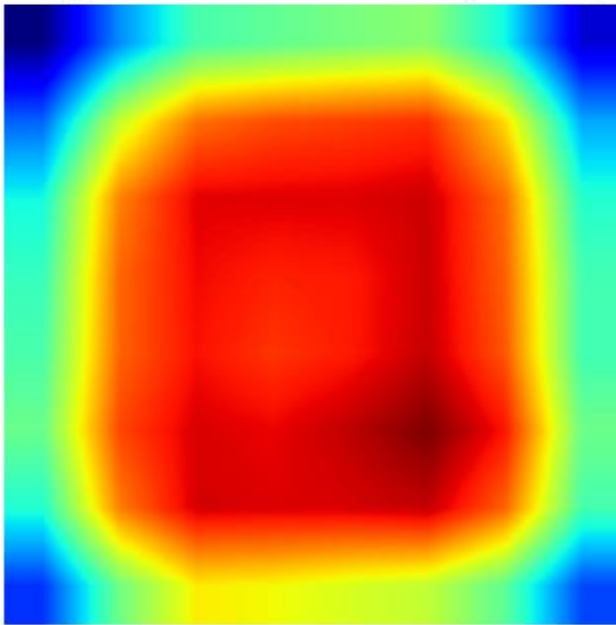
Grad-CAM Analysis (Individual Images)



The Grad-CAM visualizations on individual satellite images show that the model's attention is predominantly distributed across **broader neighborhood regions rather than individual rooftops**. High-activation areas consistently align with **environmental and infrastructural elements** such as **tree cover, open green spaces, road corridors, and spatial layout patterns**, indicating that the CNN captures **neighborhood-level context** instead of property-specific structures. **From a financial perspective**, these visual cues act as **proxies for environmental quality, accessibility, and urban organization**, all of which strongly influence **perceived desirability and market value**. As a result, the image-based model is able to **correct systematic pricing errors** in the tabular predictor by incorporating **contextual information that is difficult to quantify using structured features alone**.

Average Grad-CAM Attention (Aggregated Across Images)

Average Grad-CAM Attention (50 Images)



The **average Grad-CAM heatmap**, computed across multiple satellite images, reveals a **stable and consistent attention pattern**, with the strongest activation concentrated around the **central neighborhood region** of each image and gradually diminishing toward the edges. This indicates that the model consistently prioritizes the **immediate surroundings of each property**, rather than relying on peripheral or image-specific artifacts. Financially, this consistency implies that satellite imagery contributes **systematic and repeatable information about neighborhood quality**, supporting its role as a **reliable residual correction signal** that improves price prediction **without introducing instability or overfitting**.

Property differences between high and low value houses.





Key financial insights:

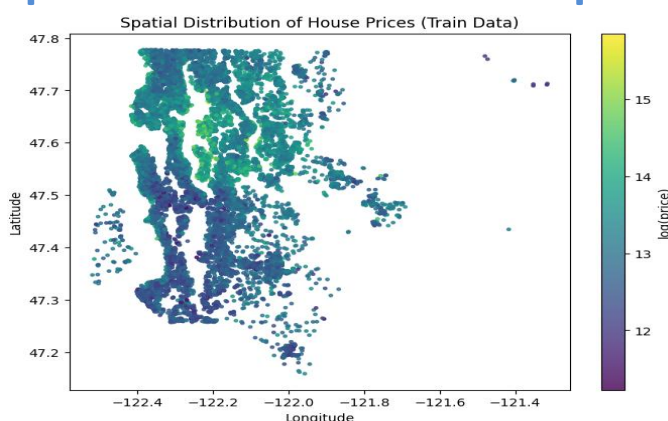
* first image corresponds to low and next one corresponds to high

Spatial openness distinguishes premium neighborhoods: High-value areas exhibit larger plots and wider spacing between houses, whereas low-price neighborhoods appear visually dense and congested. This reflects differences in privacy, livability, and land utilization that translate into higher valuations.

Road layout and accessibility influence pricing: High-priced properties are embedded within wider and more organized road networks, while lower-priced regions show narrower and irregular connectivity. Grad-CAM highlighting of road corridors indicates accessibility and infrastructure quality contribute to property value.

Planned neighborhood structure signals investment stability: High-price areas display uniform housing layouts and structured planning, while low-price regions exhibit irregular patterns and mixed land use. Consistent Grad-CAM attention over planned blocks suggests urban organization is associated with predictable appreciation.

Spatial Distribution of Property Prices

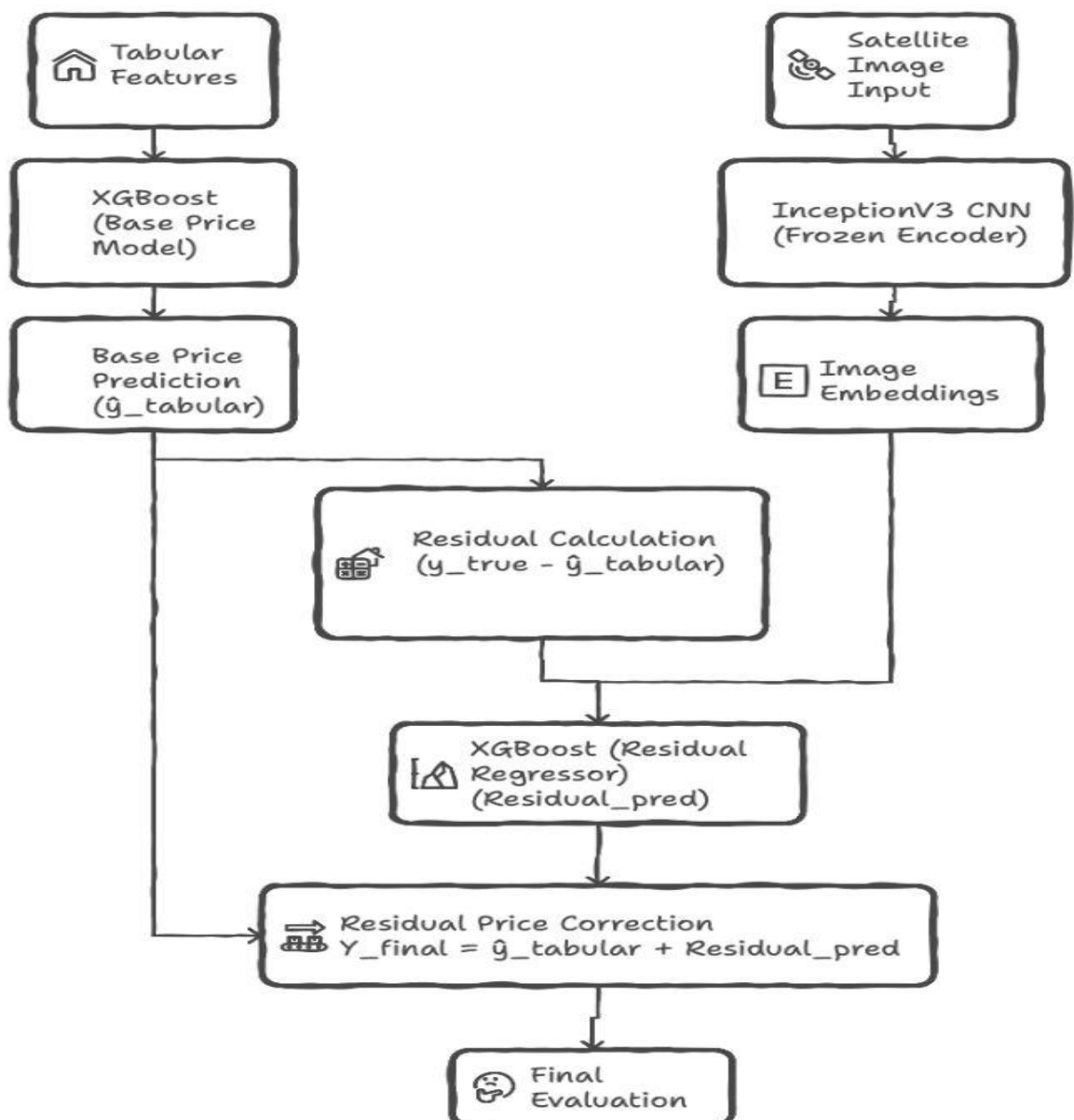


The spatial scatter plot reveals strong geographic clustering of house

prices, with contiguous pockets of high-value properties concentrated in specific corridors rather than being uniformly distributed. Higher-priced homes tend to cluster in well-defined regions, while lower-priced properties dominate peripheral and fragmented areas. This spatial structure indicates that location-driven neighborhood effects play a significant role in price formation beyond individual property attributes. Financially, such clustering reflects differences in access to amenities, infrastructure quality, environmental conditions, and local development patterns, reinforcing the importance of spatial context in real-estate valuation.

Architecture Diagram

Residual Multimodal Real Estate Price Prediction Model



Results

Model	RMSE (test)	R ² (test)
Tabular Only (XGBoost)	101,921	0.9143
Tabular + Image Embeddings	125,330	0.8704
Wealth-Aware Multimodal	117,538	0.886
Tabular + Image Residual Fusion	101,652	0.9148

The **tabular-only XGBoost model** establishes a strong baseline, **achieving an R² of 0.9143**, indicating that structured housing attributes explain the majority of price variation. **Naïve early fusion of image embeddings with tabular features degrades** validation and test performance despite improved training metrics, **highlighting overfitting and the limited usefulness of unstructured visual signals when directly concatenated**. Introducing semantic structure through a **wealth-aware image classifier improves performance over early fusion by providing coarse socioeconomic** context rather than raw embeddings. The best results are obtained using a residual learning strategy, where satellite imagery is used to predict and correct the tabular model's residual errors. **This final model improves test performance to R² = 0.9148 with low RMSE**, demonstrating that **satellite images contribute complementary neighborhood-level information while preserving the stability of the strong tabular baseline**.

Final Conclusion

Satellite imagery adds value mainly in areas where tabular features struggle to explain price differences, **by capturing neighborhood-level context such as greenery, road layout, and overall spatial structure**. Grad-CAM analysis shows that the CNN consistently focuses on these **meaningful environmental regions rather than random image areas**.